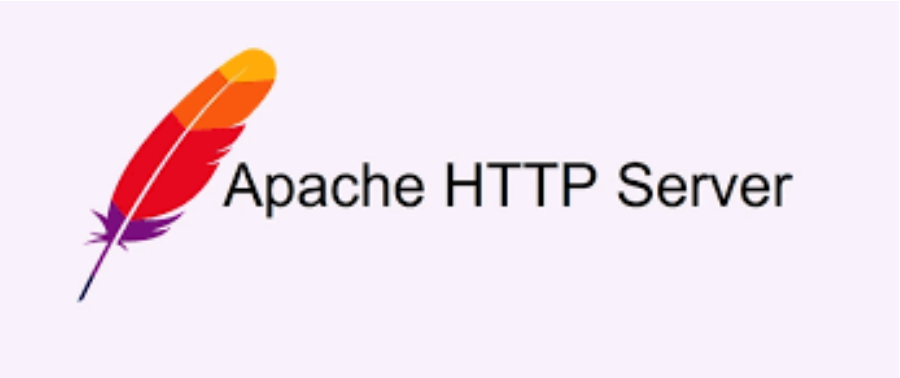
DA2D1E

INSTALACIÓN DE APACHE WEB SERVER

JESÚS DEBÉN RODRÍGUEZ



Índice

INSTALACIÓN DE APACHE WEB SERVER 3
Instalación de Apache 3
Comprobación de los cambios realizados en el sistema 3
Verificar que Apache está corriendo, desde el mismo servidor 4
Petición con el comando telnet 4
Petición con el comando wget 4
Verificar que Apache está corriendo, desde un cliente externo 4
Modificar la página por defecto de Apache 5
CREACIÓN DE SITIOS VIRTUALES 5
Configurar nombre DNS en Azure, y recopilar información para configuración DNS 5
Crear directorios para los nuevos sitios 6
Crear y personalizar ficheros de configuración de los servidores virtuales 6
Fichero ports.conf 6
Fichero port-9009.conf 7
Fichero jesusdr-da2d1e.conf 7
Fichero jesusdr-da2d1e-port-7007 8
Probar los sitios virtuales 8
Sitio por defecto 8
Sitio por el puerto 9009 8
Sitio por el nombre (jesusdr.da2d1e.es y puerto 80) 9
Sitio por el nombre y el puerto 7007 9
Sitio por el nombre (www.jesusdr.da2d1e.es y puerto 80) 9
DESPLIEGUE APLICACIÓN PHP 10
Instalación de módulos necesarios 10
Instalación de php 10
Instalación de libapache2-mod-php 10
Instalación php-sqlite3 10
Comprobación de que PHP está habilitado 11
Creación de directorios y asignación de propietarios y permisos 11

Listado del contenido de /var/www 11
Listado del contenido de /var/data 11
Copia de la aplicación al servidor 12
Creación de un nuevo sitio virtual, y pruebas del sitio 12
Posible depuración 12
PROTECCIÓN HTTPS 13
Obtención del certificado y configuración de Apache 13
Verificación de la configuración 14
Contenido del nuevo fichero creado por certbot 14
Modificaciones introducidas en el fichero ya creado previamente 15
Redirección y nueva petición HTTPS 15
Comando *systemctl list-timers | grep certbot* 15
Comando *cat certbot.timer* 16
Comando *systemctl status certbot.timer* 16
Comando *systemctl cat certbot.service* 16
Comando *sudo certbot renew --dry-run* 17

INSTALACIÓN DE APACHE WEB SERVER

Instalación de Apache

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ sudo service apache2 status
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Tue 2025-11-04 16:31:08 UTC; 1min 52s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 95434 (apache2)
      Tasks: 55 (limit: 449)
     Memory: 5.3M (peak: 8.0M)
        CPU: 44ms
     CGroup: /system.slice/apache2.service
             ├─95434 /usr/sbin/apache2 -k start
             ├─95437 /usr/sbin/apache2 -k start
             └─95438 /usr/sbin/apache2 -k start

Nov 04 16:31:08 JDR-Ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 04 16:31:08 JDR-Ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
jesusdr@JDR-Ubuntu:~$ sudo ss -tlnp
State  Recv-Q Send-Q Local Address:Port  Peer Address:Port Process
LISTEN 0      4096   127.0.0.53%lo:53         0.0.0.0:*     users:(("systemd-resolve",pid=521,fd=15))
LISTEN 0      4096        0.0.0.0:22         0.0.0.0:*     users:(("sshd",pid=90149,fd=3),("systemd",pid=1,fd=215))
LISTEN 0      4096      127.0.0.54:53         0.0.0.0:*     users:(("systemd-resolve",pid=521,fd=17))
LISTEN 0      4096            [::]:22            [::]:*     users:(("sshd",pid=90149,fd=4),("systemd",pid=1,fd=216))
LISTEN 0      511                *:80              *:*     users:(("apache2",pid=95438,fd=4),("apache2",pid=95437,fd
=4),("apache2",pid=95434,fd=4))
jesusdr@JDR-Ubuntu:~$
```

Comprobación de los cambios realizados en el sistema

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ cat /etc/passwd | grep www
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
jesusdr@JDR-Ubuntu:~$ cat /etc/group | grep www
www-data:x:33:
jesusdr@JDR-Ubuntu:~$ ls -l /var/www
total 4
drwxr-xr-x 2 root root 4096 Nov  4 16:31 html
jesusdr@JDR-Ubuntu:~$ ls -l /etc/apache2/apache2.conf
-rw-r--r-- 1 root root 7178 Aug 11 11:10 /etc/apache2/apache2.conf
jesusdr@JDR-Ubuntu:~$ ls -l /etc/apache2/ports.conf
-rw-r--r-- 1 root root 274 Mar 18  2024 /etc/apache2/ports.conf
jesusdr@JDR-Ubuntu:~$ ls -l /etc/apache2/mods-available | head -n 5
total 568
-rw-r--r-- 1 root root  100 Mar 18  2024 access_compat.load
-rw-r--r-- 1 root root  331 Mar 18  2024 actions.conf
-rw-r--r-- 1 root root   66 Mar 18  2024 actions.load
-rw-r--r-- 1 root root  743 Mar 18  2024 alias.conf
jesusdr@JDR-Ubuntu:~$ ls -l /etc/apache2/mods-enabled | head -n 5
total 0
lrwxrwxrwx 1 root root 36 Nov  4 16:31 access_compat.load -> ../mods-available/access_compat.load
lrwxrwxrwx 1 root root 28 Nov  4 16:31 alias.conf -> ../mods-available/alias.conf
lrwxrwxrwx 1 root root 28 Nov  4 16:31 alias.load -> ../mods-available/alias.load
lrwxrwxrwx 1 root root 33 Nov  4 16:31 auth_basic.load -> ../mods-available/auth_basic.load
jesusdr@JDR-Ubuntu:~$ ls -l /etc/apache2/sites-available
total 12
-rw-r--r-- 1 root root 1286 Mar 18  2024 000-default.conf
-rw-r--r-- 1 root root 4573 Mar 18  2024 default-ssl.conf
jesusdr@JDR-Ubuntu:~$ ls -l /etc/apache2/sites-enabled
total 0
lrwxrwxrwx 1 root root 35 Nov  4 16:31 000-default.conf -> ../sites-available/000-default.conf
jesusdr@JDR-Ubuntu:~$ ls -l /etc/apache2/conf-available | head -n 5
total 20
-rw-r--r-- 1 root root  269 Mar 18  2024 charset.conf
-rw-r--r-- 1 root root 3178 Mar 18  2024 localized-error-pages.conf
-rw-r--r-- 1 root root  143 Mar 18  2024 other-vhosts-access-log.conf
-rw-r--r-- 1 root root 1827 Mar 18  2024 security.conf
jesusdr@JDR-Ubuntu:~$ ls -l /etc/apache2/conf-enabled | head -n 5
total 0
lrwxrwxrwx 1 root root 30 Nov  4 16:31 charset.conf -> ../conf-available/charset.conf
lrwxrwxrwx 1 root root 44 Nov  4 16:31 localized-error-pages.conf -> ../conf-available/localized-error-pages.conf
lrwxrwxrwx 1 root root 46 Nov  4 16:31 other-vhosts-access-log.conf -> ../conf-available/other-vhosts-access-log.conf
lrwxrwxrwx 1 root root 31 Nov  4 16:31 security.conf -> ../conf-available/security.conf
jesusdr@JDR-Ubuntu:~$
```

Verificar que Apache está corriendo, desde el mismo servidor

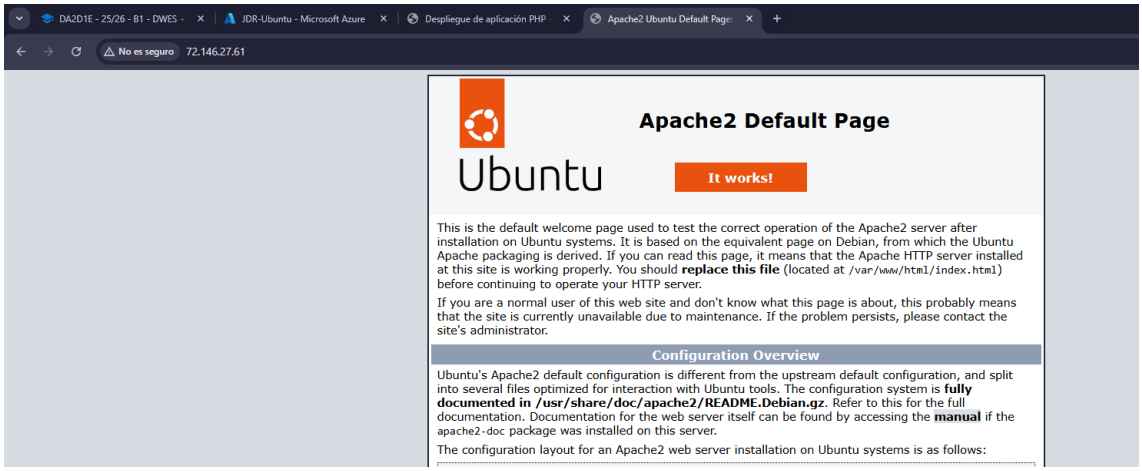
Petición con el comando telnet

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
HEAD / HTTP/1.1
HOST:127.0.0.1

HTTP/1.1 200 OK
Date: Wed, 05 Nov 2025 18:54:23 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Wed, 05 Nov 2025 18:42:52 GMT
ETag: "29c7-642dd50e94e2a"
Accept-Ranges: bytes
Content-Length: 10695
Vary: Accept-Encoding
Content-Type: text/html

Connection closed by foreign host.
jesusdr@JDR-Ubuntu:~$
```

Petición con el comando wget

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ wget http://localhost
--2025-11-05 18:54:56--  http://localhost/
Resolving localhost (localhost)... 127.0.0.1
Connecting to localhost (localhost)|127.0.0.1|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10695 (10K) [text/html]
Saving to: ‘index.html.1’

index.html.1                100%[===================================================>]  10.44K  --.-KB/s    in 0s

2025-11-05 18:54:56 (1.07 GB/s) - ‘index.html.1’ saved [10695/10695]

jesusdr@JDR-Ubuntu:~$
```

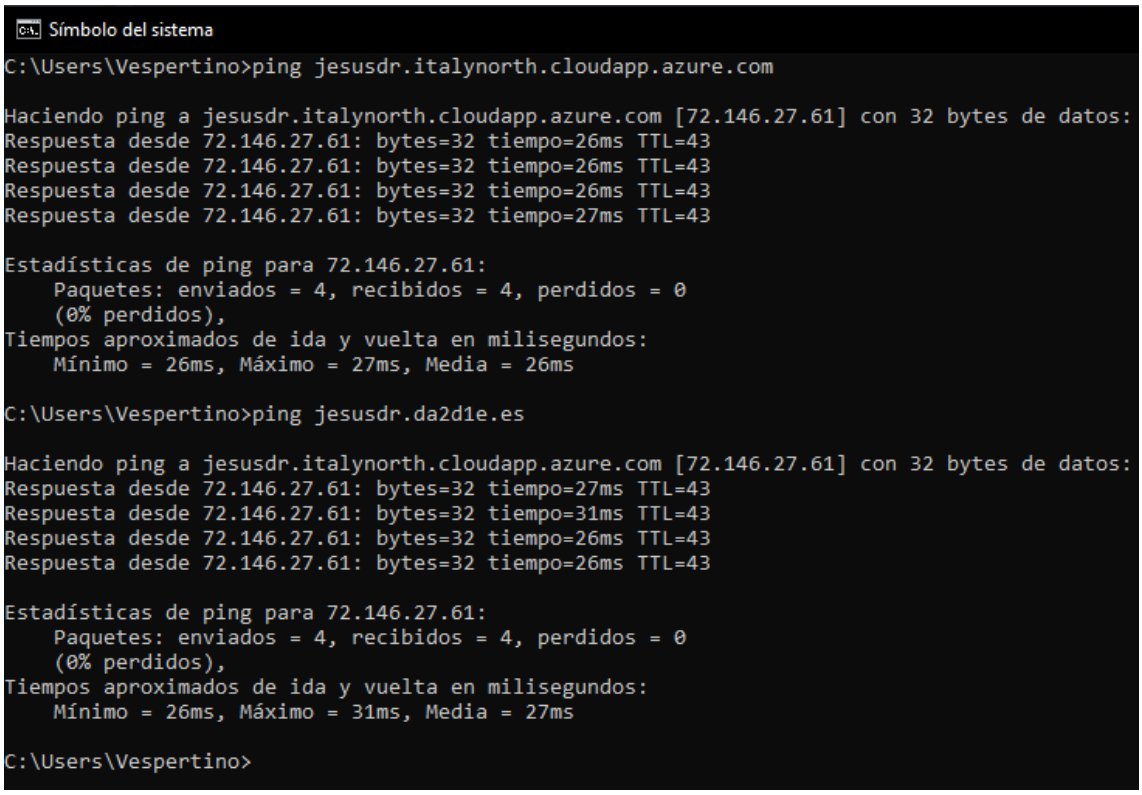
Verificar que Apache está corriendo, desde un cliente externo



Modificar la página por defecto de Apache

CREACIÓN DE SITIOS VIRTUALES

Configurar nombre DNS en Azure, y recopilar información para configuración DNS



```
Símbolo del sistema
C:\Users\Vespertino>ping jesusdr.italynorth.cloudapp.azure.com

Haciendo ping a jesusdr.italynorth.cloudapp.azure.com [72.146.27.61] con 32 bytes de datos:
Respuesta desde 72.146.27.61: bytes=32 tiempo=26ms TTL=43
Respuesta desde 72.146.27.61: bytes=32 tiempo=26ms TTL=43
Respuesta desde 72.146.27.61: bytes=32 tiempo=26ms TTL=43
Respuesta desde 72.146.27.61: bytes=32 tiempo=27ms TTL=43

Estadísticas de ping para 72.146.27.61:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 26ms, Máximo = 27ms, Media = 26ms

C:\Users\Vespertino>ping jesusdr.da2d1e.es

Haciendo ping a jesusdr.italynorth.cloudapp.azure.com [72.146.27.61] con 32 bytes de datos:
Respuesta desde 72.146.27.61: bytes=32 tiempo=27ms TTL=43
Respuesta desde 72.146.27.61: bytes=32 tiempo=31ms TTL=43
Respuesta desde 72.146.27.61: bytes=32 tiempo=26ms TTL=43
Respuesta desde 72.146.27.61: bytes=32 tiempo=26ms TTL=43

Estadísticas de ping para 72.146.27.61:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 26ms, Máximo = 31ms, Media = 27ms

C:\Users\Vespertino>
```

Crear directorios para los nuevos sitios

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ ls -al /var/www
total 24
drwxr-xr-x  6 root root 4096 Nov  6 17:25 .
drwxr-xr-x 14 root root 4096 Nov  4 16:31 ..
drwxr-xr-x  2 root root 4096 Nov  5 18:58 html
drwxr-xr-x  2 root root 4096 Nov  6 16:53 jesusdr-da2d1e
drwxr-xr-x  2 root root 4096 Nov  6 16:54 jesusdr-da2d1e-port-7007
drwxr-xr-x  2 root root 4096 Nov  6 16:53 port-9009
jesusdr@JDR-Ubuntu:~$
```

Crear y personalizar ficheros de configuración de los nuevos servidores virtuales

Fichero ports.conf

```
jesusdr@JDR-Ubuntu: ~
  GNU nano 7.2                                  /etc/apache2/ports.conf
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 80
Listen 7007
Listen 9009

<IfModule ssl_module>
        Listen 443
</IfModule>

<IfModule mod_gnutls.c>
        Listen 443
</IfModule>
```

Fichero port-9009.conf

```
jesusdr@JDR-Ubuntu: ~
  GNU nano 7.2                    /etc/apache2/sites-available/port-9009.conf *
<VirtualHost *:9009>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerAdmin jesus.deben@educa.madrid.org
        DocumentRoot /var/www/port-9009

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/port-9009-error.log
        CustomLog ${APACHE_LOG_DIR}/port-9009-access.log combined
```

Fichero jesusdr-da2d1e.conf

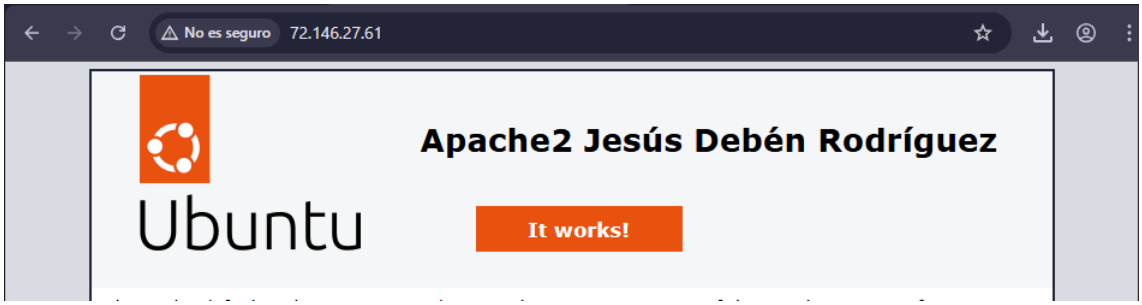
```
jesusdr@JDR-Ubuntu: ~
  GNU nano 7.2                    /etc/apache2/sites-available/jesusdr-da2d1e.conf
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerAdmin jesus.deben@educa.madrid.org
        ServerName jesusdr.da2d1e.es
        ServerAlias www.jesusdr.da2d1e.es
        DocumentRoot /var/www/jesusdr-da2d1e

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/jesusdr-da2d1e-error.log
        CustomLog ${APACHE_LOG_DIR}/jesusdr-da2d1e-access.log combined
```

Fichero jesusdr-da2d1e-port-7007

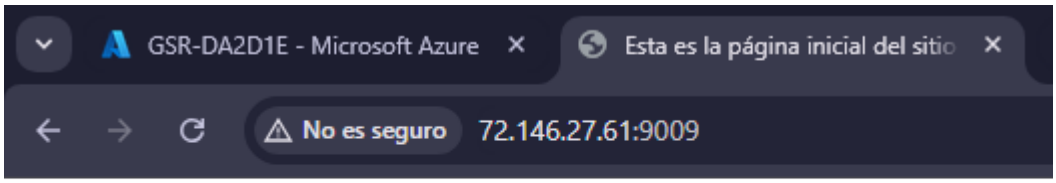
```
jesusdr@JDR-Ubuntu: ~
  GNU nano 7.2                    /etc/apache2/sites-available/jesusdr-da2d1e-port-7007.conf
<VirtualHost *:7007>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerAdmin jesus.deben@educa.madrid.org
        ServerName jesusdr.da2d1e.es
        ServerAlias www.jesusdr.da2d1e.es
        DocumentRoot /var/www/jesusdr-da2d1e-port-7007

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/jesusdr-da2d1e-port-7007-error.log
        CustomLog ${APACHE_LOG_DIR}/jesusdr-da2d1e-port-7007-access.log combined
```

Probar los sitios virtuales

Sitio por defecto

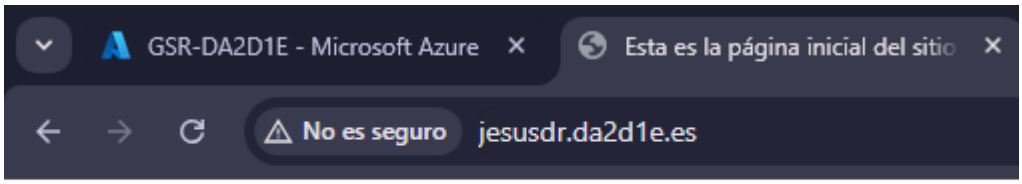


Sitio por el puerto 9009



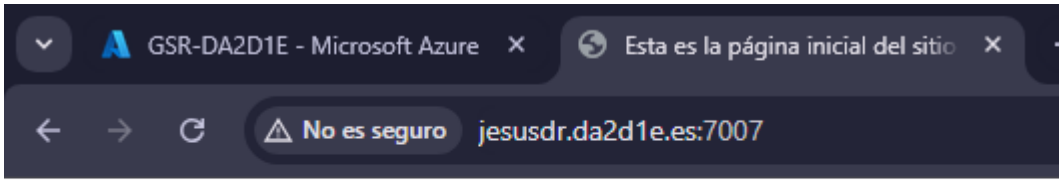
Esta es la página inicial del sitio port-9009

Sitio por el nombre (jesusdr.da2d1e.es y puerto 80)



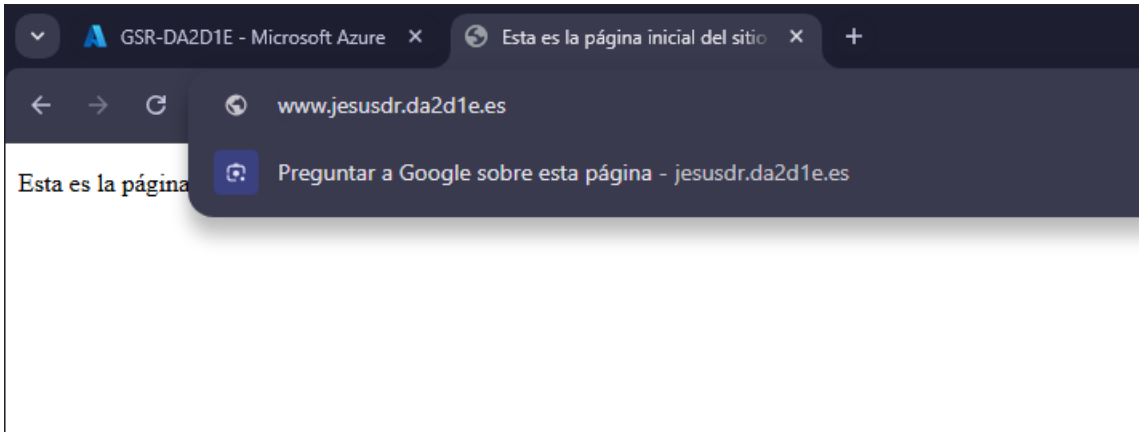
Esta es la página inicial del sitio jesusdr-da2d1e

Sitio por el nombre y el puerto 7007



Esta es la página inicial del sitio jesusdr-da2d1e-port-7007

Sitio por el nombre (www.jesusdr.da2d1e.es y puerto 80)



DESPLIEGUE APLICACIÓN PHP

Instalación de módulos necesarios

Instalación de php

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ sudo apt install php
[sudo] password for jesusdr:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php8.3 php php-common php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-readline
0 upgraded, 8 newly installed, 0 to remove and 15 not upgraded.
Need to get 4918 kB of archives.
After this operation, 22.4 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 php-common all 2:93ubuntu2 [13.9 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-common amd64 8.3.6-0ubuntu0.24.04.5 [740 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.5 [371 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.5 [13.5 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-cli amd64 8.3.6-0ubuntu0.24.04.5 [1915 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapache2-mod-php8.3 amd64 8.3.6-0ubuntu0.24.04.5 [1851 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3 all 8.3.6-0ubuntu0.24.04.5 [9174 B]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 php all 2:8.3+93ubuntu2 [4076 B]
```

Instalación de libapache2-mod-php

```
jesusdr@JDR-Ubuntu: ~

No VM guests are running outdated hypervisor (qemu) binaries on this host.
jesusdr@JDR-Ubuntu:~$ sudo apt install libapache2-mod-php
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  libapache2-mod-php
0 upgraded, 1 newly installed, 0 to remove and 15 not upgraded.
Need to get 4224 B of archives.
After this operation, 15.4 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libapache2-mod-php all 2:8.3+93ubuntu2 [4224 B]
Fetched 4224 B in 0s (65.4 kB/s)
Selecting previously unselected package libapache2-mod-php.
(Reading database ... 97360 files and directories currently installed.)
Preparing to unpack .../libapache2-mod-php_2%3a8.3+93ubuntu2_all.deb ...
Unpacking libapache2-mod-php (2:8.3+93ubuntu2) ...
Setting up libapache2-mod-php (2:8.3+93ubuntu2) ...
Scanning processes...
Scanning linux images...
```

Instalación php-sqlite3

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ sudo apt install php-sqlite3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  php8.3-sqlite3
The following NEW packages will be installed:
  php-sqlite3 php8.3-sqlite3
0 upgraded, 2 newly installed, 0 to remove and 15 not upgraded.
Need to get 36.1 kB of archives.
After this operation, 159 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-sqlite3 amd64 8.3.6-0ubuntu0.24.04.5 [34.2
Get:2 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 php-sqlite3 all 2:8.3+93ubuntu2 [1844 B]
Fetched 36.1 kB in 0s (326 kB/s)
Selecting previously unselected package php8.3-sqlite3.
(Reading database ... 97363 files and directories currently installed.)
Preparing to unpack .../php8.3-sqlite3_8.3.6-0ubuntu0.24.04.5_amd64.deb ...
Unpacking php8.3-sqlite3 (8.3.6-0ubuntu0.24.04.5) ...
```

Comprobación de que PHP está habilitado

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ ls -al /etc/apache2/mods-enabled/
total 8
drwxr-xr-x 2 root root 4096 Nov 11 15:20 .
drwxr-xr-x 8 root root 4096 Nov  6 18:03 ..
lrwxrwxrwx 1 root root   36 Nov  4 16:31 access_compat.load -> ../mods-available/access_compat.load
lrwxrwxrwx 1 root root   28 Nov  4 16:31 alias.conf -> ../mods-available/alias.conf
lrwxrwxrwx 1 root root   28 Nov  4 16:31 alias.load -> ../mods-available/alias.load
lrwxrwxrwx 1 root root   33 Nov  4 16:31 auth_basic.load -> ../mods-available/auth_basic.load
lrwxrwxrwx 1 root root   33 Nov  4 16:31 authn_core.load -> ../mods-available/authn_core.load
lrwxrwxrwx 1 root root   33 Nov  4 16:31 authn_file.load -> ../mods-available/authn_file.load
lrwxrwxrwx 1 root root   33 Nov  4 16:31 authz_core.load -> ../mods-available/authz_core.load
lrwxrwxrwx 1 root root   33 Nov  4 16:31 authz_host.load -> ../mods-available/authz_host.load
lrwxrwxrwx 1 root root   33 Nov  4 16:31 authz_user.load -> ../mods-available/authz_user.load
lrwxrwxrwx 1 root root   32 Nov  4 16:31 autoindex.conf -> ../mods-available/autoindex.conf
lrwxrwxrwx 1 root root   32 Nov  4 16:31 autoindex.load -> ../mods-available/autoindex.load
lrwxrwxrwx 1 root root   30 Nov  4 16:31 deflate.conf -> ../mods-available/deflate.conf
lrwxrwxrwx 1 root root   30 Nov  4 16:31 deflate.load -> ../mods-available/deflate.load
lrwxrwxrwx 1 root root   26 Nov  4 16:31 dir.conf -> ../mods-available/dir.conf
lrwxrwxrwx 1 root root   26 Nov  4 16:31 dir.load -> ../mods-available/dir.load
lrwxrwxrwx 1 root root   26 Nov  4 16:31 env.load -> ../mods-available/env.load
lrwxrwxrwx 1 root root   29 Nov  4 16:31 filter.load -> ../mods-available/filter.load
lrwxrwxrwx 1 root root   27 Nov  4 16:31 mime.conf -> ../mods-available/mime.conf
lrwxrwxrwx 1 root root   27 Nov  4 16:31 mime.load -> ../mods-available/mime.load
lrwxrwxrwx 1 root root   34 Nov 11 15:20 mpm_prefork.conf -> ../mods-available/mpm_prefork.conf
lrwxrwxrwx 1 root root   34 Nov 11 15:20 mpm_prefork.load -> ../mods-available/mpm_prefork.load
lrwxrwxrwx 1 root root   34 Nov  4 16:31 negotiation.conf -> ../mods-available/negotiation.conf
lrwxrwxrwx 1 root root   34 Nov  4 16:31 negotiation.load -> ../mods-available/negotiation.load
lrwxrwxrwx 1 root root   29 Nov 11 15:20 php8.3.conf -> ../mods-available/php8.3.conf
lrwxrwxrwx 1 root root   29 Nov 11 15:20 php8.3.load -> ../mods-available/php8.3.load
lrwxrwxrwx 1 root root   33 Nov  4 16:31 reqtimeout.conf -> ../mods-available/reqtimeout.conf
lrwxrwxrwx 1 root root   33 Nov  4 16:31 reqtimeout.load -> ../mods-available/reqtimeout.load
lrwxrwxrwx 1 root root   31 Nov  4 16:31 setenvif.conf -> ../mods-available/setenvif.conf
lrwxrwxrwx 1 root root   31 Nov  4 16:31 setenvif.load -> ../mods-available/setenvif.load
lrwxrwxrwx 1 root root   29 Nov  4 16:31 status.conf -> ../mods-available/status.conf
lrwxrwxrwx 1 root root   29 Nov  4 16:31 status.load -> ../mods-available/status.load
```

Creación de directorios y asignación de propietarios y permisos

Listado del contenido de /var/www

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ ls -al /var/www
total 28
drwxr-xr-x  7 root root 4096 Nov 11 15:29 .
drwxr-xr-x 15 root root 4096 Nov 11 15:32 ..
drwxr-xr-x  2 root root 4096 Nov 11 15:29 eventos-app
drwxr-xr-x  2 root root 4096 Nov  5 18:58 html
drwxr-xr-x  2 root root 4096 Nov  6 17:30 jesusdr-da2d1e
drwxr-xr-x  2 root root 4096 Nov  6 17:31 jesusdr-da2d1e-port-7007
drwxr-xr-x  2 root root 4096 Nov  6 17:29 port-9009
jesusdr@JDR-Ubuntu:~$
```

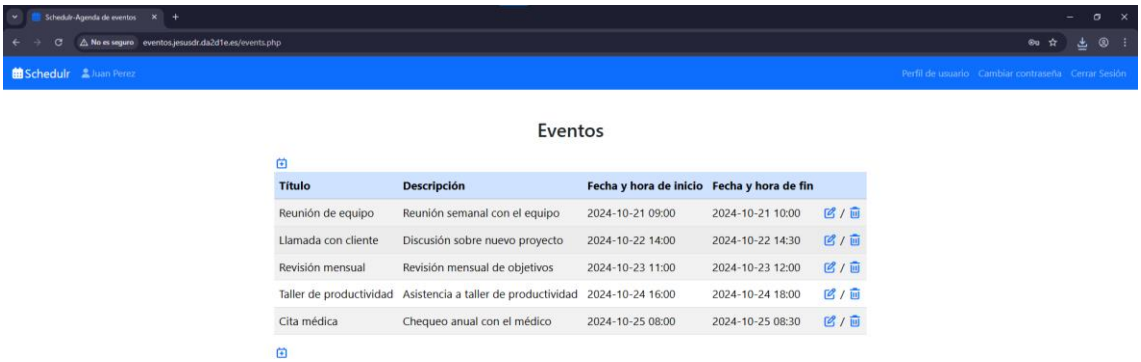
Listado del contenido de /var/data

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ ls -al /var/data
total 12
drwxr-xr-x  3 root     root     4096 Nov 11 15:36 .
drwxr-xr-x 15 root     root     4096 Nov 11 15:32 ..
drwxr-xr-x  2 www-data www-data 4096 Nov 11 15:33 eventos-app
jesusdr@JDR-Ubuntu:~$
```

Copia de la aplicación al servidor

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ jesusdr@JDR-Ubuntu:~$ sudo ls -al /var/www/eventos-app/aplicacion/
total 68
drwx------ 2 root root 4096 Nov 11 16:04 .
drwxr-xr-x 4 root root 4096 Nov 11 16:04 ..
-rw-r--r-- 1 root root 2781 Nov 11 16:04 cabecera.php
-rw-r--r-- 1 root root 3042 Nov 11 16:04 delete-event.php
-rw-r--r-- 1 root root 7374 Nov 11 16:04 edit-event.php
-rw-r--r-- 1 root root 3787 Nov 11 16:04 events.php
-rw-r--r-- 1 root root  965 Nov 11 16:04 icono-aplicacion.svg
-rw-r--r-- 1 root root 4894 Nov 11 16:04 index.php
-rw-r--r-- 1 root root 2429 Nov 11 16:04 logout.php
-rw-r--r-- 1 root root 5563 Nov 11 16:04 new-event.php
-rw-r--r-- 1 root root 1969 Nov 11 16:04 profile.php
-rw-r--r-- 1 root root 9224 Nov 11 16:04 register.php
jesusdr@JDR-Ubuntu:~$
```

Creación de un nuevo sitio virtual, y pruebas del sitio



Posible depuración

- Error de la dirección de la BBDD: modificar dentro del servidor en cada página necesaria la dirección con la nueva dirección '*/var/data/eventos-app/calendar.db*'.
- Error de permisos de los ficheros de la aplicación: modificar los permisos con el comando '*chmod -R 755 /var/www/eventos-app*'.
- Error con el propietario de la carpeta *data* y sus ficheros: modificar el propietario con el comando '*chmod -R www-data:www-data /var/data*'.
- Error con alguna dirección de enlaces: modificarla a la correcta, teniendo en cuenta que tienen que ser direcciones relativas.

PROTECCIÓN HTTPS

Obtención del certificado y configuración de Apache

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ sudo certbot --apache
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Enter email address (used for urgent renewal and security notices)
 (Enter 'c' to cancel): jesus.deben@educa.madrid.org

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Please read the Terms of Service at
https://letsencrypt.org/documents/LE-SA-v1.5-February-24-2025.pdf. You must
agree in order to register with the ACME server. Do you agree?
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
(Y)es/(N)o: y

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Would you be willing, once your first certificate is successfully issued, to
share your email address with the Electronic Frontier Foundation, a founding
partner of the Let's Encrypt project and the non-profit organization that
develops Certbot? We'd like to send you email about our work encrypting the web,
EFF news, campaigns, and ways to support digital freedom.
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
(Y)es/(N)o: y
Account registered.

Which names would you like to activate HTTPS for?
We recommend selecting either all domains, or all domains in a VirtualHost/server block.
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
1: jesusdr.da2d1e.es
2: eventos.jesusdr.da2d1e.es
3: www.jesusdr.da2d1e.es
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Select the appropriate numbers separated by commas and/or spaces, or leave input
blank to select all options shown (Enter 'c' to cancel): 2
Requesting a certificate for eventos.jesusdr.da2d1e.es

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/eventos.jesusdr.da2d1e.es/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/eventos.jesusdr.da2d1e.es/privkey.pem
This certificate expires on 2026-02-11.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for eventos.jesusdr.da2d1e.es to /etc/apache2/sites-available/eventos.jesusdr.da2d1
e-le-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://eventos.jesusdr.da2d1e.es

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

Verificación de la configuración

Contenido del nuevo fichero creado por certbot

```
jesusdr@JDR-Ubuntu: /etc/apache2/sites-available
  GNU nano 7.2                          eventos.jesusdr.da2d1e-le-ssl.conf
<IfModule mod_ssl.c>
<VirtualHost *:443>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        ServerName eventos.jesusdr.da2d1e.es

        ServerAdmin jesus.deben@educa.madrid.org
        DocumentRoot /var/www/eventos-app/aplicacion

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/eventos-app-error.log
        CustomLog ${APACHE_LOG_DIR}/eventos-app-access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
RewriteEngine on
# Some rewrite rules in this file were disabled on your HTTPS site,
# because they have the potential to create redirection loops.

# RewriteCond %{SERVER_NAME} =eventos.jesusdr.da2d1e.es
# RewriteRule ^ https://%{SERVER_NAME}%{REQUEST_URI} [END,NE,R=permanent]

SSLCertificateFile /etc/letsencrypt/live/eventos.jesusdr.da2d1e.es-0001/fullchain.pem
SSLCertificateKeyFile /etc/letsencrypt/live/eventos.jesusdr.da2d1e.es-0001/privkey.pem
Include /etc/letsencrypt/options-ssl-apache.conf
</VirtualHost>
</IfModule>
```

Modificaciones introducidas en el fichero ya creado previamente

```
jesusdr@JDR-Ubuntu: /etc/apache2/sites-available
  GNU nano 7.2                                  eventos.jesusdr.da2d1e.conf
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        ServerName eventos.jesusdr.da2d1e.es

        ServerAdmin jesus.deben@educa.madrid.org
        DocumentRoot /var/www/eventos-app/aplicacion

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/eventos-app-error.log
        CustomLog ${APACHE_LOG_DIR}/eventos-app-access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
RewriteEngine on
RewriteCond %{SERVER_NAME} =eventos.jesusdr.da2d1e.es
RewriteRule ^ https://%{SERVER_NAME}%{REQUEST_URI} [END,NE,R=permanent]
</VirtualHost>
```

Redirección y nueva petición HTTPS

Elements | Console | Sources | Network | Performance | Memory | Application | Lighthouse

Preserve log | Disable cache | No throttling

Filter | Invert | More filters | All | Fetch/XHR | Doc | CSS | JS | Font | Img | Media | Manifest | Socket | Wasm | Other

10 ms | 20 ms | 30 ms | 40 ms | 50 ms | 60 ms | 70 ms | 80 ms | 90 ms | 100 ms | 110 ms | 120 ms | 130 ms

Name	Status	Type	Initiator	Size	Time
eventos.jesusdr.da2d1e.es	301	document / Redirect	Other	(disk cache)	4 ms
eventos.jesusdr.da2d1e.es	200	document	eventos.jesusdr.da2d1e.es/	1.3 kB	88 ms
bootstrap.min.css	200	stylesheet	(index):9	(disk cache)	12 ms
all.min.css	200	stylesheet	(index):10	(disk cache)	12 ms
fa-solid-900.woff2	200	font	all.min.css	(memory cache)	0 ms

Comando *systemctl list-timers | grep certbot*

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ systemctl list-timers | grep certbot
Fri 2025-11-14 07:42:30 UTC      14h -                              - certbot.timer                certbot.service
jesusdr@JDR-Ubuntu:~$
```

Comando *cat certbot.timer*

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ systemctl cat certbot.timer
# /usr/lib/systemd/system/certbot.timer
[Unit]
Description=Run certbot twice daily

[Timer]
OnCalendar=*-*-* 00,12:00:00
RandomizedDelaySec=43200
Persistent=true

[Install]
WantedBy=timers.target
jesusdr@JDR-Ubuntu:~$
```

Comando *systemctl status certbot.timer*

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ systemctl status certbot.timer
● certbot.timer - Run certbot twice daily
     Loaded: loaded (/usr/lib/systemd/system/certbot.timer; enabled; preset: enabled)
     Active: active (waiting) since Thu 2025-11-13 16:30:43 UTC; 1h 1min ago
    Trigger: Fri 2025-11-14 07:42:30 UTC; 14h left
   Triggers: ● certbot.service

Nov 13 16:30:43 JDR-Ubuntu systemd[1]: Started certbot.timer - Run certbot twice daily.
jesusdr@JDR-Ubuntu:~$
```

Comando *systemctl cat certbot.service*

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ systemctl cat certbot.service
# /usr/lib/systemd/system/certbot.service
[Unit]
Description=Certbot
Documentation=file:///usr/share/doc/python-certbot-doc/html/index.html
Documentation=https://certbot.eff.org/docs
[Service]
Type=oneshot
ExecStart=/usr/bin/certbot -q renew --no-random-sleep-on-renew
PrivateTmp=true
jesusdr@JDR-Ubuntu:~$
```

Comando *sudo certbot renew --dry-run*

```
jesusdr@JDR-Ubuntu: ~
jesusdr@JDR-Ubuntu:~$ sudo certbot renew --dry-run
Saving debug log to /var/log/letsencrypt/letsencrypt.log

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Processing /etc/letsencrypt/renewal/eventos.jesusdr.da2d1e.es-0001.conf
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Simulating renewal of an existing certificate for eventos.jesusdr.da2d1e.es

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Processing /etc/letsencrypt/renewal/eventos.jesusdr.da2d1e.es.conf
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Renewal configuration file /etc/letsencrypt/renewal/eventos.jesusdr.da2d1e.es.conf is broken.
The error was: expected /etc/letsencrypt/live/eventos.jesusdr.da2d1e.es/cert.pem to be a symlink
Skipping.

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Congratulations, all simulated renewals succeeded:
  /etc/letsencrypt/live/eventos.jesusdr.da2d1e.es-0001/fullchain.pem (success)

Additionally, the following renewal configurations were invalid:
  /etc/letsencrypt/renewal/eventos.jesusdr.da2d1e.es.conf (parsefail)
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
0 renew failure(s), 1 parse failure(s)
Ask for help or search for solutions at https://community.letsencrypt.org. See the logfile /var/log/letsencrypt/letse
ncrypt.log or re-run Certbot with -v for more details.
jesusdr@JDR-Ubuntu:~$
```